Bags, Vectors & Transformers

Foundations of Word Embeddings · Day 2, Part 1

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Welcome Back — Where We Left Off

Yesterday we built the **classic toolkit** for turning text into data. A quick recap:

What we covered

- Text → tokens → clean tokens
- The **document-term matrix**
- TF-IDF weighting
- Dictionary, supervised ML, topic models

And where it broke

- Word order lost
- Synonyms look unrelated
- Sparse, high-dimensional
- No sense of context or meaning

Today we address that last column — starting with the idea of representing meaning itself.

Before We Dive In

Any questions from yesterday?

Anything from the notebooks, the document-term matrix, the three approaches, or the limitations we saw?

Now is a good moment — today builds directly on these ideas.

The Problem We Want to Solve

Bag-of-words treats every word as an **isolated symbol**. “happy”, “joyful”, and “concrete” are just three different columns — equally unrelated to one another.

What we *want* is a representation where a word’s **meaning** is captured, so that words with similar meanings have similar representations. The question is:

Where could “meaning” possibly come from, if all we have is raw text?

The answer is one of the most influential ideas in language science.

The Distributional Hypothesis

> *"You shall know a word by the company it keeps."*
>
> — J.R. Firth, 1957

The **distributional hypothesis** is the idea that words appearing in **similar contexts** tend to have **similar meanings**. Meaning is not an intrinsic label — it can be inferred from **patterns of use**.

This is a genuinely powerful move: it turns an abstract, philosophical question ("what is meaning?") into something **measurable** from text. If we can measure the contexts a word appears in, we can estimate its meaning.

An Intuition: Guess the Word

Suppose you keep seeing an unfamiliar word — let's call it **"wug"** — in sentences like these:

"I poured myself a cold glass of **wug**."
"She drinks **wug** with breakfast every morning."
"The carton of **wug** in the fridge has gone sour."

What is a wug? Call out your guess.

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❓ What is a wug? Call out your guess.

You have never been told what "wug" means — but you already have a strong guess. Something like **milk**. How? You inferred it **entirely from context**: the words that surround it (glass, drinks, breakfast, carton, fridge, sour).

That is the distributional hypothesis at work — and it is exactly what embedding methods automate over billions of sentences.

Guess the Hidden Word: Round 1

YOUR TURN The same word is hidden in every sentence. Call out your guess after each one: does it change?

1 The _____ were brought inside before the storm.

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COWS Each new context rules out alternatives: children, then goats, then cows. Embedding models do the same thing, over thousands of contexts per word.

Guess the Hidden Word: Round 2

YOUR TURN Same game. This one should feel familiar.

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1. The second _____ wanted a completely different analysis.
2. We replied to every point the _____ raised.

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reviewer A twist: GloVe's nearest neighbors of "reviewer" are *allmusic*, *ign*, *booklist* and *kirkus*. Trained on Wikipedia and news, GloVe thinks a reviewer writes about albums, games and books. Context depends on the corpus.

From Contexts to Vectors

If meaning comes from context, we can represent each word by **the company it keeps** — as a **vector of numbers**. Words used in similar contexts get similar vectors.

Bag-of-words vector

- One dimension **per vocabulary word**
- Tens of thousands of dimensions
- Mostly zeros (sparse)
- Dimensions are **just word identities**

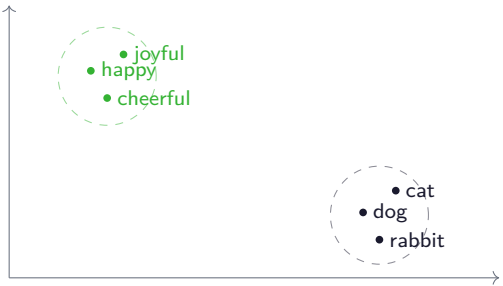
Embedding vector

- A few **hundred** dimensions
- **Dense** (mostly non-zero)
- Learned from context
- Dimensions capture **latent features** of meaning

*We call these dense vectors **word embeddings**: each word is "embedded" as a point in a shared meaning-space.*

Words as Points in Space

The key payoff: in embedding space, **distance means something**. Words with similar meanings sit **close together**; unrelated words sit far apart.



"happy", "joyful", "cheerful" cluster together; "dog", "cat", "rabbit" form a separate cluster. The model was never *told* these groupings — it discovered them from context. *(A 2-D sketch; real embeddings have hundreds of dimensions.)*

Which Word Is Closest to *coffee*?

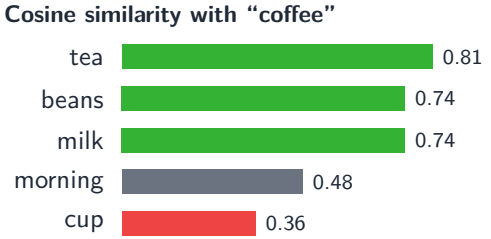
YOUR TURN In GloVe, the embeddings you'll use in the notebook, which of these words is **most** similar to "coffee"? And which is **least** similar?

- cup
- tea
- morning
- milk
- beans

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"cup" comes last, although "a cup of coffee" is everywhere. Words end up close when they fit the *same slots* ("a hot _____ with breakfast"), not when they sit *next to* each other: similar means roughly **substitutable**.

A Famous Consequence: Analogies

Because meaning is encoded geometrically, **directions** in the space can carry meaning too. The classic example:

$$\textbf{king} - \textbf{man} + \textbf{woman} \approx \textbf{queen}$$

The vector that takes you from "man" to "woman" is roughly the *same* vector that takes you from "king" to "queen". A "gender direction" emerges in the space, along with directions for tense, plurality, even country-to-capital.

This was the result that made embeddings famous around 2013 — and it falls out of the distributional hypothesis, no hand-coding required.

Complete the Analogy

YOUR TURN What would *you* answer? Then we look at GloVe's top answer (the same model as in the notebook).

amsterdam : netherlands :: **brussels** : ?

tokyo : japan :: **nairobi** : ?

big : bigger :: **small** : ?

walking : walked :: **swimming** : ?

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walking : walked ::	**swimming** : ?	**athletics** ✗

Two out of four. The famous examples are the ones that *work*; plenty of analogies don't. Treat analogies as an illustration of the geometry, not as a measurement.

Two Flavors of Embeddings

Static embeddings

· One fixed vector **per word**

· Word2Vec, GloVe

· "bank" has **one** vector, no matter the sentence

· Simple, fast, interpretable

We start here.

Contextual embeddings

· A vector **per word-in-context**

· BERT and friends

· "bank" gets **different** vectors in different sentences

· Powerful, but heavier

We build to this later.

Both rest on the same foundation: meaning from context. The difference is whether context is baked in once, or computed fresh each time.

Your Word, Your Context

TURN TO YOUR NEIGHBOR 2 minutes

Think of a word that matters in your research, and whose meaning depends on **who** uses it or **where**. For example "positive" in health, "conservative" in politics, "sustainable" in agriculture.

- Which words would **surround** it in *your* data?
- Would its neighbors be the same in **general news** or Wikipedia?

Keep your word in mind: on the next slide you can look it up right away.

Look It Up: The Embedding Playground



denisejroth.github.io/bags-vectors-transformers/playground

YOUR TURN Scan with your phone camera. No login needed.

The same GloVe vectors as in the notebook, in your browser:

- **nearest neighbors** of any word
- **compare** two words
- **analogies**
- **two poles**: closer to “she” or “he”?
- **odd one out**

Look up the word you just discussed. Is its neighborhood what you expected?

Foundations: The Takeaways

- ✔ Bag-of-words fails because it treats words as **isolated symbols**
- ✔ The **distributional hypothesis**: a word's meaning can be inferred from the **contexts it appears in**
- ✔ **Embeddings** represent each word as a **dense vector** in a shared meaning-space
- ✔ In that space, **distance** reflects similarity and **directions** can carry meaning (analogies)
- ✔ Two flavors: **static** (one vector per word) and **contextual** (one vector per use)

Next: how do we actually *learn* these vectors from data? We open up Word2Vec and GloVe — and then get our hands on real embeddings in Python.